Slavery, Causation, and Contribution

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Abstract

I use a metaphysical orthodoxy to argue for a heterodox account of egalitarian justice. I do so via a debate from outside of philosophy. The debate is about whether enslaved labor contributed to America's prosperity: historians say yes, economists say no. I argue that they are talking past one another. The historians understand contribution in terms of causation; the economists understand it in terms of counterfactual dependence. It is a metaphysical orthodoxy that counterfactual dependence is not necessary for at least one central kind of causation: you can cause an outcome without making a difference to its occurrence, for example when *preempting* a would-be replacement. This orthodoxy allows for a re-interpretation of a principle that partly animates the slavery debate but which now lies mostly in the ash heap of the history of political philosophy—the principle of reward according to contribution. The received view is that this principle would vindicate serious inequality. But if we understand contribution in terms of this central kind of causation, then the received view can be flipped on its head.

This paper is about two ideas, both with a long history. The first idea has lived its life entirely outside of philosophy. It is an empirical claim, the claim that enslaved labor made America rich. The second idea has likewise lived its life in the outdoors, but it has played an important role in political philosophy, too. It is the normative claim that workers are entitled to the fruits of their labor, that they deserve pay commensurate with the value of their productive contribution.

I argue for two theses, one about each idea. The first thesis is that the debate about whether enslaved labor made America rich—which is ongoing—is

based in part on a failure to make the philosophically orthodox distinction between counterfactual dependence and (at least one central kind of) causation. The second thesis is that this same distinction helps transform our understanding of the principle of reward according to contribution. In doing so it enables a heterodox *pro tanto* case for an egalitarian distribution of labor income.

Let me say a little more about each thesis.

The debate about whether enslaved labor made America rich is nearly as old as American slavery itself: slavery's apologists argued that it did; conventional wisdom in the North was that it did not.[1] In its most recent iteration, the "economics-of-slavery culture wars,"[2] it is a debate between history departments and economics departments—between narrative economic historians and "cliometricians," or economic historians who use quantitative methods, including those of econometrics. Over the last fifteen years, in work referred to as the "New History of Capitalism,"[3] the narrative historians have argued that enslaved labor made an important contribution to America's economic growth. In several reviews of this body of work, the cliometricians claim to have already shown, in the decades preceding the New History, that slavery if anything slowed growth, that it was an economic albatross as well as a moral catastrophe.[4]

My first thesis is that the narrative economic historians (from now on just "historians") and the cliometricians are partly talking past one another. The historians can be interpreted as talking about causation: they claim that enslaved labor was an important cause of America's prosperity. The cliometricians are talking about counterfactual dependence: they claim that America's prosperity did not depend on enslaved labor, that America would have been just as rich, if not richer, had it abolished slavery upon its founding. The causal claim and the counterfactual dependence claim are compatible. Counterfactual dependence is not necessary for (at least one central kind of) causation: a cause will make no difference to what it nonetheless causes when it *preempts* a would-be replacement. As I will explain, this is part of what is

[1] Wright 2020: 378; Wright 2022: 140–42.

[2] Wright 2017.

[3] Johnson 2013; Beckert 2014; Baptist 2014; Schermerhorn 2015; Beckert and Rockman 2016. And see, more recently, Rockman 2024.

[4] Engerman 2017; Hilt 2017; Olmstead and Rhode 2018; Majewski 2019; Wright 2020, 2022. For cliometric work written since these reviews, see Bleakley and Rhode 2024a, 2024b; Allen et al. 2025.

going on in the case of American slavery.

My second thesis is that understanding productive contribution in causal terms, rather than in terms of counterfactual dependence, helps transform the principle of reward according to contribution. Start with the following puzzle. Despite its centrality to historical socialism[5] and its continual deployment within the actual labor movement, the principle—call it the Contribution Principle—is unpopular among contemporary egalitarian political philosophers. Why? Well, it is one thing to say, in the familiar way, that some nineteenth-century factory owner in Industrial Revolution England exploits his workers because he takes too much of the fruits of their labor. But if that presupposes that workers are entitled to those fruits, then this left-wing condemnation of capitalist exploitation is going to end up vindicating serious inequality *within* the labor market. Some workers, after all, produce more fruit than others, especially now that the Industrial Revolution is behind us. This is just a version of the challenge that G. A. Cohen raised for Marxists in *Self-Ownership, Freedom, and Equality*. As it happens I think that Cohen's original challenge is misleading.[6] But even the best version of

[5] It was the clarion call of the "Ricardian" socialists: see Hall 1849 (1805): 94–104, 207; Hodgskin 1825: esp. 1, 25, 1827: 16–32, 108–9; Bray 1839: 30–3, 37–52; Gray 1826: esp. 11–18; and, more ambiguously, Thompson 1824: 145–9, 1827: 14. It is also associated with Lassalleanism, despite the fact that Lassalle did not say that much about it (1879 [1863]: 19–20; cf. 1884 [1862]). The association is presumably due to Marx's criticism in "Critique of the Gotha Program," where he famously claimed that the Contribution Principle would be left behind in the transition from socialism to communism (Marx 1978 [1875]: 529–30). This claim notwithstanding, Marxists and fellow travelers continued to invoke the principle (perhaps because they conflated it with the labor theory of value, on which see Jackson 2007: 48–9). For example, the British Labour Party's understood commitment to socialism took the form of promising, in Clause IV of its Constitution, "[t]o secure for the workers by hand or by brain the full fruits of their industry and the most equitable distribution thereof." (This text lasted until Tony Blair: Adams 1999: 144–5.)

[6] Cohen (1995: ch. 6) argues that the Marxist theory of exploitation presupposes the thesis of *self-ownership*, the most important part of which, for present purposes, is the right to transfer one's labor power, and the derivative right (Vallentyne et al. 2005: 204, n. 6) to what one gets in exchange, sometimes called the "right to income." While Cohen does not claim that anyone who thinks workers are exploited is likewise committed to the thesis of self-ownership (1995: 149), he gives the impression that anyone who thinks workers are exploited on the basis of something like the Contribution Principle is so committed. But self-ownership is neither necessary nor sufficient for the Contribution Principle; all the principle needs is the weaker idea that workers can *take normative credit* for what they achieve using their labor. Self-ownership is not sufficient because to say that a worker is entitled to the pay promised them in a labor contract—entitled to the income that they

that challenge, I here argue, can be met: if we understand contribution in causal terms, the Contribution Principle can be egalitarian. Thus egalitarian political philosophers can—and I think should—be like many egalitarians in the outside world: they can be *productivists*, people who believe that facts about who produced what help answer the question of who should get what.[7]

I will move back and forth between my two theses. In Section 1, I make a simple observation about the contribution made by low-wage workers, taken together—an observation that seems to have an egalitarian upshot. In Section 2, in order to determine how to interpret that observation, I turn to the debate about the contribution that enslaved labor made to the American economy. I argue that the debate is partly dissolved by the distinction between causation and counterfactual dependence. (I make this same argument more precisely, using a structural equations model, in the Appendix.) In Section 3, I explain that this distinction, and in particular the irrelevance of replaceability to causation, underlies the simple observation from the first section. In Section 4, I complicate the simple observation in the face of an objection but argue that its egalitarianism survives the complications.

Two preliminary notes. First, I said just above that egalitarian political philosophers can and should be productivists. But I argue only for the "can," in that my second thesis conditional: if workers deserve reward according to their contributions, then what they deserve is much more equal than is usually supposed. This thesis is worth caring about despite its being conditional because its condition is worth taking seriously. It has the distinction of having animated at least a century of socialist thought and then (more intermittently) another century or so of capitalist thought.[8] And, outside of

can get from the labor that they are entitled to transfer—is not to say that this income is deserved on the basis of what they contribute. Self-ownership blesses a contract that pays a worker to sit around doing nothing; the Contribution Principle does not. (Cf. Perry 1997.) Nor is self-ownership necessary. Consider someone who: accepts the principle and so thinks that the aforementioned worker's income should be taxed away; and, because they reject self-ownership, thinks there is no even *pro tanto* liberty-based reason not to do so. (Cf. Cohen 1995: ch. 6, n. 5 and 1979: n. 21. And note the similarity with David Gauthier's defense of taxing away "rents" [1986: 276].)

[7] I take this term from Aas 2019: 71. Cf. Berkey's (2026) use of "productive justice" and Hsieh's (2008) of "justice in production."

[8] 112 years separate the earliest Ricardian socialists from Sidney and Beatrice Webb's drafting of Clause IV for the British Labour Party's constitution—see note 5. 111 years separate John Bates Clark's contribution-based defense of his own marginalist theory of distribution (Clark 1899) from Greg Mankiw's (in)famous "Reflections on Joe the Plumber"

philosophy, it remains popular.[9]

Second, this paper is written with four audiences in mind: political philosophers, metaphysicians, economists, and historians. I think that parts of the paper will seem tedious to each group. But there is something to be gained from bringing these discussions together.

1 A Simple Observation

Imagine that all the workers in the bottom half of the global income distribution—"low-wage workers"—stopped working. What would happen? Here is a plausible sounding answer: the global economy would grind to a halt; output would fall to zero, or near enough.

According to the most popular conception of "productive contribution," a worker's contribution is the difference that they make to the value of output—at least a very rough proxy, the thought goes, for the difference they make to the satisfaction of consumer preferences—something in the neighborhood of what economists call their *marginal revenue product*.[10] Given anything like this conception of contribution, the above answer implies that the contribution of low-wage workers taken together is roughly equal to all of global

(2010). In the middle of this span, the "just deserts" interpretation of the marginalist theory of distribution helped give the "Cambridge Controversy" its strange intensity: see Robinson 1964: 57–60; Solow 1963: 10; Blaug 1975: 75; Cohen and Harcourt 2003: 210. Without meaning to imply that this just deserts interpretation is popular among contemporary economists, it at least lives on in the implied non-instrumental normative importance of "Pigouvian" exploitation, i.e., paying workers less than their marginal revenue product (Pigou 2017 [1952]: part III, ch. XIV, §3; Appendix III, §30.)

[9] See Miller 1999: ch. 4; Karagözoğlu 2012; Mulligan 2018: ch. 3.

[10] The normative claim that workers should—non-instrumentally—be rewarded according to marginal product is roughly as old as the empirical claim that under certain conditions this is what a free market for labor will do: see the aforementioned Clark 1899. (Rawls challenges this orthodox interpretation of Clark, citing a discussion by Clark's son in Spiegel 1952: Rawls 1999: 271, n. 33. But in that very discussion the younger Clark wrote that his father "had formulated his generalized marginal-productivity theory of distribution, and had satisfied himself that, when it is working properly, the wage system meets the requirements of economic justice." Spiegel 1952: 602.) For contemporary defenses, see Miller 1999: ch. 6 and Mulligan 2018. For an overview, see Behrle 2025: secs. 1.3–1.4. This paper is concerned with *output-based* conceptions of contribution—conceptions that can be thought of as identifying the "fruits" of a worker's labor. This is in part because it is these conceptions, rather than input-based ones (e.g., effort, labor time), that seem to vindicate serious inequality. For relevant discussion, see Mulligan 2018: 127–31.

output.

But the same is presumably true of the other half of the global income distribution, the high-wage workers. Without them, output would likewise fall to zero, or near enough. In this sense the two halves of the global economy contribute roughly equally. This is what I will call, in what follows, "the simple observation."

The simple observation prompts some immediate questions. Philosophers, for example, may be wondering what we need to read into the antecedents of the relevant counterfactuals ("If they stopped working…") in order to vindicate them. Economists may be wondering—and this will turn out to be related—whether these are claims about the short term or the long term.[11] Everyone may be wondering why we are drawing our line down the middle of the income distribution, as opposed to somewhere else. I will explain, later on, how I think the simple observation needs to be interpreted, and why the arbitrariness of its dividing line is not a problem. The observation is a ladder that we will kick away after climbing.

For now, note that we reach the same result as above even in an idealized global economy, a perfect free market for labor in which each worker is paid according to their marginal revenue product, per the neoclassical theory of wage determination. In such an economy, the same claims hold: the economy would seem to grind to a halt without either of its halves.

I note this because it is in an important way puzzling. If each worker is paid according to their marginal revenue product, then low-wage workers must by definition contribute—in just that sense—less than high-wage workers. How is it possible, then, that the low-wage workers as a group contribute just as much as the high-wage workers as a group? It is because in complex economies groups of workers often together contribute dramatically more than the sum of what they contribute as individuals.

This is the germ of my own argument (Behrle 2025) against the most common version of the Contribution Principle—reward according to the difference an individual makes, e.g., their marginal revenue product. I argue that this version of the principle grants workers more credit for their individual contributions than they can plausibly claim, because it is "blind" to the dependence of these contributions on what other workers "only together"

[11] They may also be wondering whether I am assuming, e.g., a Leontief production function. The answer is no: as I explain in Section 3.1, what we assume on my intended interpretation of the simple observation is not a lack of substitutability but a lack of substitution, via a strong *ceteris paribus* condition.

do, that is, on the latter's collective contributions. Put differently, reward according to individual contribution registers collective contributions, but in the wrong way: it smuggles the achievements of large groups of workers into the individual contributions that the groups' achievements make possible.

We should understand this argument as part of a long-running attempt to identify the role that other people play in each worker's contributions, and the normative significance thereof. This is itself an important theme in the history of egalitarian political thought, one that grew up alongside the principle it challenges, the Contribution Principle. As early as 1839 we find the Ricardian socialist J. F. Bray observing that the enormous value of the invention of the steam engine depends on the worker who "fills it with water and kindles the fire under it"; "Every man is a link, and an indispensable link, in the chain of effects."[12] Proudhon's *What is Property?* is published a year later. In one of its more lucid passages, Proudhon complains that capitalists, in paying their workers one by one, pay "nothing for that immense power which results from the union and harmony of laborers."[13] The theme receives its first really sustained attention from early twentieth-century British liberals and leftists, including the "Fabian" socialists, perhaps most of all L. T. Hobhouse and J. A. Hobson.[14] In contemporary political philosophy, the theme re-emerges in a famous passage in Elizabeth Anderson's "What is the Point of Equality?"—which passage I discuss at the end of the paper—and also in the work of Samuel Scheffler and T. M. Scanlon (the latter citing Hobhouse).[15] In contemporary American politics, the theme made it into one of the most infamous lines of the 2010's: Obama's "you didn't build that," which gives Sean Aas's 2019 paper on the theme its title.[16]

I make note of this history because the argument of this paper presupposes it: I assume that the individualistic version of the Contribution Principle has

[12] Bray 1839: 45.

[13] Proudhon 1994 (1840): 91. I prefer to think of this, anachronistically and against the grain of Proudhon's own example—the obelisk—as making the very general point that workers' marginal products can be lower than average product, along with an implicit objection to the resulting surplus going to the people who contribute by supplying capital that they should not privately own. For work in economics that is relevant to a narrower reading of the passage, see Alchian and Demsetz 1972: sec. III; Card et al. 2018: sec. II.

[14] For an overview, see Jackson 2007: 49–53. For Hobson's views in particular see Allett 1981: ch. 3; Pullen 2010: 61–8.

[15] Anderson 1999: 321; Scheffler 2003: 191; Scanlon 2013: 114.

[16] Aas 2019. For different developments of the "you didn't build that [alone]" idea, see Alm 2010; Von Platz 2022.

been taken off the table, if not by my own argument then by one of its antecedents.[17] In effect the question I ask is: What happens to the Contribution Principle once we take on board this challenge that grew up alongside it—once we ascend from the level of individuals to the level of collectives, as in the simple observation?

2 Slavery and Causation

I now turn to the debate about the contribution that enslaved labor made to the American economy. I am going to extract from this debate a lesson about the relationship between *replaceability* and a central kind of causation. In the next section, I will show that this lesson underlies the simple observation. To anticipate: a worker's contribution, understood in terms of this central kind of causation, is given by (roughly) the difference they make *barring their replacement*; and it is only with this condition that the low- and high-wage workers contribute equally.

2.1 Introducing the Debate

In 2011, there was a conference convened by Brown and Harvard that inaugurated the New History of Capitalism. The first sentence of Sven Beckert and Seth Rockman's introduction to the conference volume, *Slavery's Capitalism*, reads:

> During the eighty years between the American Revolution and the Civil War, slavery was indispensable to the economic development of the United States.[18]

Claims like this are central to the New History. Here is Edward Baptist, for another example, in *The Half Has Never Been Told*:

> [T]he commodification and suffering and forced labor of African Americans is what made the United States powerful and rich.[19]

[17] But for a recent critique of Anderson, Scheffler, Behrle, and Aas, see Lister forthcoming.

[18] Beckert and Rockman 2016: 1.

[19] Baptist 2014: xxi. For other examples, see Baptist 2014: xxiii, xxvi, 3–4, 33, 113, 128, 229, 311, 312, 316, 319, 350, 385, 407–8, 411, 413; Baptist 2016: 32–3, 41; Schermerhorn 2016: 211; Shankman 2016: 244; Brophy 2016: 263; Martin 2016: 119.

It is claims like these that the cliometricians take themselves to deny. That is our debate.

This debate is morally charged in a way that might seem surprising. It might seem surprising because it is not clear what normative issues turn on the question of slavery's economic consequences. The New Historians seem to have two issues in mind. The first is about capitalism: if slavery was important for capitalism's development, at least in the United States, then that might tell us something normatively important about capitalism, at least in the United States.[20] The second is about enslaved workers: if their labor made America rich, this would seem to count in favor of the claim that their descendants are owed reparations.[21] I set the first of these two ideas aside. But my proposed interpretation of the debate leaves space for the second.

Beckert and Rockman identify several mechanisms by which enslaved labor is supposed to have made the United States rich:

> [1] Slave-grown cotton was the most valuable export of the United States and [2] one of the few American-made goods that attracted specie [i.e., non-paper money] into the nation's financial system. [3] Cotton also offered a reason for entrepreneurs and inventors to build manufactories in such places as Lowell, Pawtucket, and Paterson, thereby connecting New England's Industrial Revolution to the advancing plantation frontier of the Deep South. [4] And financing cotton growing, as well as marketing and transporting the crop, was a source of great wealth for the nation's merchants and bankers.[22]

For simplicity, I focus on versions of the first and third claims, though what

[20] Thus the title of Matthew Desmond's (2019) contribution to the *New York Times'* "1619 Project": "If you want to understand the brutality of American capitalism, you have to start on the plantation."

[21] Baptist (2014: xix): "If slavery was outside of US history. . .—if indeed it was a drag and not a rocket booster to American economic growth—then slavery was not implicated in US growth, success, power, and wealth. Therefore none of the massive quantities of wealth and treasure piled by that economic growth is owed to African Americans." To anticipate what I say in Section 3.2: I do not think that this inference survives reflection. It suggests that if a lazy but talented capitalist hires a worker to produce something of value, and if in that worker's absence the capitalist would have produced it more efficiently, then the worker is owed no share of the value of what they produce. That is a bizarre view. (Thanks to Daniel Viehoff for this example.)

[22] Beckert and Rockman 2016: 26.

I say applies to the others. In particular, I focus on these simple versions of the first and third claims:

COTTON Enslaved labor was important for the production of cotton, which was very valuable.

INDUSTRY Slave-grown cotton was important for the production of cotton textiles in New England, which was important for America's industrialization.

Let me say more about what I do not discuss. I confine myself to the United States. So I do not discuss the claim, from then-future Trinidadian leader Eric Williams' 1944 *Capitalism and Slavery*, that slavery was essential to the Industrial Revolution in England.[23] Nor do I discuss the possible dependence of America's economic development on slavery in places outside of the United States (e.g., via New England's trade with the West Indian "sugar islands"[24]). In addition, the debate that I focus on—about whether enslaved labor made America rich—is distinct from the debate about whether slave-based cotton production was "efficient," the modern iteration of which was initiated by Robert Fogel and Stanley Engerman's controversial 1974 book *Time on the Cross*.[25] It is neither necessary nor sufficient for the claim that enslaved labor made America rich that slave-based cotton production was efficient in any of the senses at play in that debate.[26]

[23] There is a large literature about this, the first half of the "Williams thesis." See, e.g., Darity 1982; Solow and Engerman 1987; Findlay 1990; Findlay and O'Rourke 2007, ch. 6; Inikori 2002; Acemoglu et al. 2005; Blackburn 2011, ch. 4; and Hudson 2014. Wright 2020 helpfully contrasts cliometricians' attitudes toward the first half of the Williams thesis with their attitudes toward the New Historians' analogous claim about American economic development.

[24] Bailey 1986, 1990; Kimball 2016. If the first half of the Williams thesis is true, then American economic development also depended on enslaved labor in the West Indies to whatever extent it depended on the Industrial Revolution in England, as Wright (2006: 123) notes.

[25] To get a sense of why many of Fogel and Engerman's claims were rejected at the time, see David et al. 1976. For recent evidence that bears on the debate, see Bleakley and Rhode 2024a and 2024b.

[26] Regarding necessity: Suppose that (a) slave-based cotton production was less efficient (in any sense) than "free labor"–based cotton production would have been, but that (b) free labor could not have been induced to produce cotton in the United States before 1861. (See generally the distinction between slavery as a production process and as a

Back to the debate at hand. The cliometricians seem to deny COTTON and INDUSTRY. Here is Gavin Wright:

> Did slavery play a primary and indispensable role in the rise of the US economy to world preeminence? This proposition has deep historical roots: Pro-slavery apologists argued that "slavery was the nursing mother of the prosperity of the North".... Karl Marx wrote in 1846 that "without slavery there would be no cotton, without cotton there would be no modern industry".... More recently, claims about the centrality of slavery to nineteenth-century US economic growth have featured in the national conversation on race, perhaps most notably in an intellectual insurgency known as the New History of Capitalism, some of which has filtered into the popular press through channels like the Pulitzer Prize-winning "1619 Project" published by the *New York Times* (2019). Yet this proposition has been rejected by virtually every economic historian [i.e., cliometrician] who has examined the issue.[27]

Why do the cliometricians deny that slavery was indispensable for the production of cotton? Because they believe that, in the absence of slavery, cotton would have been produced in the South by non-enslaved workers—"free labor"—just like it was after the Civil War.[28] Indeed Wright argues that a counterfactual free-labor-based South would have produced *more* cotton,

system of property rights in Wright 2006: 20–7.) Regarding sufficiency: flip the situation. Suppose that (c) slave-based production was more efficient but (d) total cotton production would have been higher with a free labor force, because that labor would have been more *elastically* supplied (see below)—so much higher that, via knock-on effects, output would have been greater even per capita.

[27] Wright 2022: 123–4.

[28] Commenting on Beckert's acknowledgment that American cotton production rebounded after the Civil War, Wright writes, "He does not seem to notice that [this acknowledgment] undermine[s] the previous 300 pages of his book" (Wright 2020: 372). (See also Majewski 2019 and Olmstead and Rhode 2018: 6.) But the rebounding of cotton production after the Civil War is not enough to show that production earlier in the century did not counterfactually depend on slavery, which is what Wright takes Beckert to be claiming in the book. Cotton was produced after the Civil War in part by former slaves. It is open to Beckert to observe that the size of the free labor force produced by the emancipation of an enslaved labor force is not a good guide to the size of the free labor force that would have been assembled earlier via immigration. (Elster [1978: 208]: "If you step in and measure the importance of the ladder at the point where it has become possible

in part because the supply of enslaved labor was relatively inelastic after the closing of the Atlantic slave trade in 1807.[29]

Why do the cliometricians deny that slave-grown cotton was indispensable for cotton textile production? One reason follows from the previous: if enslaved labor was not necessary for domestic cotton production, then it was not necessary for the cotton textiles that used domestic cotton. But Wright suggests, in addition, that in the absence of domestic cotton New England's mills simply would have sourced it from elsewhere.[30] John Majewski (2019) argues, separately, that the production of cotton textiles was itself not necessary for the North's industrialization: he rejects the historians' implicit "leading-sector" hypothesis.

The cliometricians also argue that slavery was in other ways actively bad for the American economy, either because: the economic inequality it generated led the South to develop various bad institutions;[31] and/or it crowded out investment in physical capital, like factories and machines;[32] and/or it stood in the way of the kind of ongoing technological innovation that pow-

to kick it away, you should not be surprised if it turns out to be negligible.") Indeed this is one of the main ideas in Beckert's book—the idea that, before certain infrastructural, technological, and legal changes, slavery was the only mechanism for "the mobilization of very large numbers of workers on very short notice" (91; see also Findlay and O'Rourke 2007: 342 and Findlay 1990: 14). I find this idea difficult to evaluate, because we never get a canonical list of the relevant changes. In any case, we have to read Wright's criticism of Beckert in light of Wright's other work, which other work suggests that he is skeptical of this claim; see his discussion of the North's superiority in attracting immigrants in Wright 1978: ch. 4 and Wright 2006: 32–4, 57–8 (and see the next sentence in the main text).

[29] Wright 2020: 370. In addition, slave-based Southern farms plausibly underspecialized in cotton, growing food alongside it. This was because they were trying to expropriate the maximum value from enslaved laborers, which meant coming up with other agricultural work to be done during cotton production's seasonal nadirs. In the jargon, slavery made agriculture more of a *fixed-cost* enterprise. Wright 2020: 370; Gallman 1970; Anderson and Gallman 1977; Lindstrom 1970. Cf. North 1961: 67–8, 101–2, 128–9, 140–1.

[30] "As a bulky but lightweight commodity, raw cotton travels easily, and transportation costs play little if any role in textiles geography" (Wright 2020: 375). One might worry that the claim is a little quick, at least for the period after the cotton tariff was lowered in 1846, when American prices for American and British cotton textiles were similar (Harley 1992, table 1).

[31] This is the "Engerman-Sokoloff Hypothesis." See Engerman and Sokoloff 1997. Cf. Nunn 2008; Acemoglu et al. 2008.

[32] Ransom and Sutch 1988, esp. 139–41; Wright 2006: 61. For related points, see the second thesis of Wright 1978: ch. 4.

ered the "Schumpeterian" North;[33] and/or it reduced the size of the manufacturing and service-sector workforces.[34] I ignore these claims of negative contribution for a moment, but I will come back to them.

So, to restate, the cliometricians argue that (i) slavery was not indispensable for cotton production and that (ii) slave-based domestic cotton production was not indispensable for the production of cotton textiles, or industrialization. In this sense they seem to deny COTTON and INDUSTRY. And they argue that (iii) slavery otherwise held economic development back. From all of this the cliometricians conclude that, had the United States abolished slavery earlier, it would have become just as rich, if not richer.

I want to argue that the cliometricians and the New Historians are to some extent talking past one another. I do not deny the three cliometric claims just mentioned, nor the concluding inference, which I am in any case not qualified to do. I suggest instead that these three claims are compatible with the following interpretation of COTTON and INDUSTRY: enslaved people were partly *causally responsible* for cotton production, and for cotton textile production, and, in this way, for America's prosperity.

2.2 An Orthodoxy

This compatibility follows from an orthodoxy within philosophy, one with almost no dissenters:[35] on one important notion of causation—perhaps the most important, and the most ordinary—counterfactual dependence is not necessary for it; effects need not counterfactually depend on their causes.

This is the lesson of cases of *redundant* causation. Consider the following stock example.[36] Suzy and Billy are throwing rocks at a glass bottle. Suzy hits it; the bottle shatters. As it happens, though, had she not thrown her rock, Billy would have hit the bottle with his. Here we have causation without apparent counterfactual dependence of effect on cause: Suzy's throw caused the bottle to shatter, but Suzy does not seem to make a difference to the bottle shattering; had she not thrown the rock, it still would have shattered. In the literature, this kind of redundant causation is called *preemption*, because Suzy preempts Billy, her would-be replacement.

[33] Majewski 2019.

[34] Allen et al. 2025.

[35] But see Northcott 2021.

[36] Lewis 2000: 184. According to Hitchcock (2013: 131 n. 4), the example originated with a draft of Hall 2004.

In noting that effects need not counterfactually depend on their causes in cases like this, I am not suggesting that causation and counterfactual dependence are unrelated. Indeed counterfactual theories of causation remain popular within philosophy. But the burden of those theories is to identify some more complicated relationship between counterfactual dependence and causation precisely in the light of the fact that effects need not counterfactually depend on their causes. For example: the bottle's shattering does not depend on Suzy's throw *simpliciter*, but it does depend on Suzy's throw *holding Billy's inaction fixed*; holding this fixed "unmasks" the dependence of the effect on its actual cause.[37]

I say more about this notion of causation, and the relationship between causation and counterfactual dependence, below. But it will be helpful first to see the way in which slave-based production resembles a case of preemption.

2.3 The Debate in Light of the Orthodoxy

The New Historians will sometimes present their claims like this:

> *[I]n actual historical fact* there was no nineteenth-century capitalism without slavery.[38]

> [T]he North's forms of entrepreneurship, innovation, and market competition [invite] the counterfactual claim that the American economic takeoff could have happened without slavery. *Perhaps it might have, but the fact remains that it didn't.*[39]

Eric Hilt, a cliometrician, criticizes this focus on actuality:

> Many historians apparently have a strong distaste for counterfactual histories. . . . Yet the reason economic historians think about counterfactuals is not due to an interest in specifying alternative histories. Rather, it is because all statements about causal relationships contain counterfactuals. To say that the gold standard caused the Great Depression is to say that absent the gold standard, the Great Depression would not have happened; these two statements are equivalent.[40]

[37] Woodward 2003: 84; Weslake 2015: 8–9.
[38] Johnson 2013: 254, emphasis mine.
[39] Beckert and Rockman 2016: 3, emphasis mine.
[40] Hilt 2017: 529.

Hilt's final claim is, per the orthodoxy, false for at least one central notion of causation—the one at work in the case of Suzy and Billy. In that case, counterfactual dependence of effect on cause is not necessary for causation, and so not necessary and sufficient, and so counterfactual dependence claims are not in general equivalent to causal claims, materially or otherwise. The mooted truth of the cliometricians' counterfactual claims—that slavery was not indispensable for the production of cotton or cotton textiles—does not establish that enslaved labor did not cause America to become rich.

To make this vivid, consider these two pairs of claims. The first pair:

(a) In the first half of the nineteenth century, almost all of the cotton in the United States was produced by enslaved people.

(b) Had the United States never allowed chattel slavery, (at least as much) cotton still would have been produced.

The second pair:

(c) The production of cotton textiles in New England, which helped industrialize the North, used slave-grown cotton.

(d) If these mills hadn't used slave-grown cotton, they would have gotten cotton from elsewhere.

(a) is one disambiguation—a causal disambiguation—of COTTON. And it is consistent with (b): (b) is the denial of a distinct, counterfactual disambiguation of COTTON. Likewise, (c) is a causal disambiguation of INDUSTRY, and it is consistent with (d), which is the denial of a distinct, counterfactual disambiguation of INDUSTRY. This consistency is the hallmark of cases of preemption.

In at least this sense, then, the cliometricians' putative denial of the historians' COTTON and INDUSTRY need not be a denial at all: what they deny, à la (b) and (d), is counterfactual dependence, and such dependence is not necessary for a standard causal interpretation of the two theses, à la (a) and (c).

2.4 Two Kinds of Causation?

I have just suggested that the cliometricians are advancing counterfactual dependence claims that are, per the orthodoxy, consistent with COTTON and INDUSTRY when those theses are given a standard causal interpretation.

I now briefly consider whether these counterfactual dependence claims are themselves causal. In the course of doing so I will say a little more about how philosophers understand the kind of causation that I am reading the New Historians as invoking, and explain how this understanding illuminates additional features of the slavery debate.

Distinguish *type* causation ("smoking causes cancer") from *token* or *actual* causation ("his smoking caused his cancer").

At the level of type causation, it is common to make a further distinction between, on the one hand, *total* (or *net*) causes and *contributing* (or *component*) causes. This further distinction is often illustrated with a famous example, from Hesslow 1976: the effect of early birth control pills on thrombosis.[41] These pills both (a) increase the risk of thrombosis and (b) decrease the risk of pregnancy, which itself increases the risk of thrombosis. If the increase and decrease cancel out, there seems to be a sense in which the pills both cause and do not cause thrombosis.

Hitchcock (2001b) argues that we can make this same distinction at the level of actual causation. One of his own illustrations will be familiar to cliometricians: Fogel on railroads. Fogel (1964) claimed that railroads were less causally important to the nineteenth-century American economy than had been supposed: while railroads did reduce transportation costs relative to the options that they superseded, investment in railroads also reduced investment in competing options, like roads and canals, which would have themselves reduced transportation costs. Here is Hitchcock's graphical representation of this claim—a *directed acyclic graph*, part of a *causal model*—where the negative signs indicate the direction of effect and the thickness of the arrows their strength:

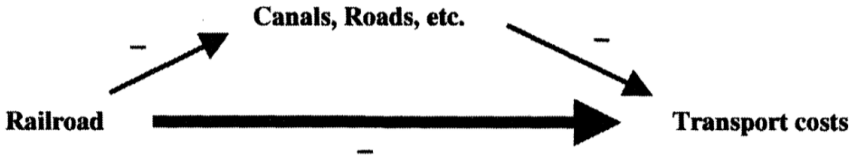


Figure 2

[42]

Railroads were both a total and a contributing cause of reduced transport costs, but their total *effect* was smaller than their contributory *effect* (along the bottom causal pathway).

[41] Pearl 2009: 127; Woodward 2003: 49–50; Hitchcock 2001b: 364.

[42] Hitchcock 2001b: 367.

Determining total effect is, conceptually, easy: we vary the candidate cause and see what, per our model, would happen. In particular, we do not hold fixed anything that is causally intermediate between it and the outcome.[43] To determine the contributory effect along one causal pathway, we need to hold more things fixed. The question of what we hold fixed, and in particular at what states or "values" (of the variables by which we represent them in our causal model), is a difficult one: this is the question to which much of the philosophical work on actual causation in this century has been addressed.[44] But in the present context we can sidestep this difficulty. The case of the railroad and the case of American slavery involve redundancy in the form of preemption. In preemption cases, we can determine the contributory effect along one causal pathway by varying the candidate cause while holding fixed at their *actual* states everything that is not intermediate between it and the effect on that very pathway..[45]

So, in the case of the railroad, we estimate total effect by asking, roughly: What would have happened to transportation costs if we hadn't expanded the railroads? We estimate contributory effect by asking, roughly: What would have happened to transportation costs if we hadn't expanded the railroads, holding fixed the actual level of investment in other forms of transportation?

Note that we can, if we like, work backwards from the notion of total and contributory effects to the notion of total and contributing causes, as the former furnish a necessary and (normally[46]) sufficient condition for each of the latter: having a positive effect of the relevant kind (or, in some contexts,

[43] I use the word "vary" in the sense of a causal *intervention*. (See, e.g., Woodward 2003: 46–8.) This means that, as with the assignment to treatment in a randomized trial, we are in effect holding fixed at their actual values events that are not causally intermediate between the cause and the outcome, including common causes of both.

[44] Important examples include Pearl 2000; Hitchcock 2001a; Woodward 2003; Halpern and Pearl 2005; Hall 2007; Hitchcock 2009; Weslake 2015; Gallow 2021. Cf. Andreas and Günther 2021.

[45] In Pearl's later terminology, the contributory effect in which we are interested in the present context is the *natural direct effect* "under the reverse transition" (Pearl et al. 2016: 121–2).

[46] Bracketing the problem of *structural* or *counterfactual isomorphs*: see Hall 2007: 44ff; Hitchcock 2007; Menzies 2017. Jäger 2026 offers counterexamples to the claim that counterfactual dependence, and *a fortiori* positive contributory effect, is sufficient for (contributory) causation even after we have helped ourselves to a *default/deviant* distinction, the normal solution to the problem of counterfactual isomorphs. But his counterexamples are recherché: they are meant to militate against defining causation in terms of counterfactual dependence but not against the sufficiency of the latter for the former in everyday cases.

any non-zero effect).

Economists will find total effect familiar: it is, in the *potential outcomes* framework that is popular in contemporary econometrics, the difference between Y_1 and Y_0—outcome with treatment and without.[47]

Philosophers might note that, when cause and outcome are binary, total causation does not seem importantly different from *non-backtracking* counterfactual dependence, and thus from the "concept" of causation that Hall (2004) dubbed "dependence." Indeed, even without binary events, we can say of a total cause with a large effect that the outcome taking on some large value, as opposed to a much smaller one, counterfactually depends on that cause.

But this highlights something else, something puzzling. Almost all contemporary philosophical work on actual causation is work on what I have called, following Pearl, Hitchcock, and Woodward, contributory or component causation. The impression one gets from this literature is that, whatever is true at the level of type causation, at the level of actual causation there is no such thing as "total" causation: all actual causes are contributing causes, and only contributing causes. As may now be obvious, this is the kind of actual causation that I have been interpreting the New Historians as invoking; and it is with an eye to the almost exclusive focus of the contemporary philosophical literature that I referred to it as "perhaps the most important, and the most ordinary" kind of causation.

Why the implied non-reality of actual total causation? Return to the case of Suzy and Billy. Suzy has no total effect on the bottle's shattering. So she is not a total cause of the bottle's shattering. This sounds very odd. How could there be any notion of causation on which Suzy is not a cause of the bottle's shattering?[48] Hitchcock suggests that pragmatics determine the naturalness of a total or contributory reading of any given causal claim. But if this were a question of pragmatics then we should be able to do something to get ourselves to hear a felicitous (total) reading, and I at least cannot.[49]

[47] That said, the potential outcomes framework is not blind to the phenomenon of preemption: preemption can show up as a violation of what Rubin (1986) dubbed the "stable-unit-treatment-value assumption" (SUTVA), at least with jerry-rigged individuations of treatment and outcome. If the treatment is "be given a rock" and the outcome whether the rock is thrown, SUTVA is violated in the case of Suzy and Billy, since the outcome of Billy receiving treatment is not independent of Suzy receiving treatment. (For distinct SUTVA-based criticism of cliometric work on slavery, see Francis 2026.)

[48] See Northcott 2021 for an attempt to bite this bullet.

[49] Hitchcock actually confines himself to the terminology of "effect." He does not speak

The claim that Suzy is not a total cause of the bottle shattering is in effect an invitation to see two contributory causal pathways as "canceling out": on the one hand Suzy causes a bottle to shatter; but on the other hand, by preempting Billy, she causes a bottle not to shatter. These cannot cancel out, however, because the second claim is false. Actual causation is partly defined by the requirement that the cause and effect actually occur. Since it is not the case that the bottle did not shatter, it is not the case that Suzy is an actual cause of its not shattering.

This problem is less likely to arise in the standard economic context. Change the case so that Billy is throwing rocks at a second bottle. But as before they are taking turns, and as before Suzy preempts him. Now consider the claim that Suzy is not a (total) cause of the disvalue of a bottle shattering. This sounds at least somewhat less odd. Here there is a bottle that did not shatter. And perhaps the fungibility of economic value clears the ground for the necessary "canceling out."

I am not sure what to think about the reality of total causation at the token level. But if we are willing to grant its reality, then we can understand the cliometric claims about slavery as compatible with many of the historians' claims without needing to deny that the former claims are themselves causal. This follows from my earlier discussion and the observation, just above, that total causation is not importantly different than non-backtracking counterfactual dependence.

Whether or not we grant this, I now want to show how even the above rough characterization of the difference between total and contributory effect illuminates further features of the slavery debate.

Olmstead and Rhode take Baptist to task for making a classic "double-counting error" when calculating the value of cotton for the American economy.[50] This is the error of adding the value of an input to the value of what is made with it. Because the value of an output already includes the value of its inputs, adding the value of its inputs separately is to double count that value. We avoid this error by subtracting, rather than adding, the value of the inputs, leaving us with "value added." But the value-added framework is ill-

of total (or, as he puts it, "net") *causes*. This allows him to say: "Since 'net effect' is not part of ordinary English, it is difficult to marshal intuitive support for [the definition of net effect]," and more generally for the felicitousness of sentences in which the property of having no net effect is predicated of a preemptor (2001b: 384). But this will not help in the present context: the cliometricians want to talk about *causes*.

[50] Baptist 2014: 321–2; Olmstead and Rhode 2018: 13. Rhode 2024.

suited for estimating a contributory causal effect, which is what I take Baptist to be gesturing toward in the relevant passage.[51] To estimate the contributory effect of enslaved labor on American prosperity along the causal pathway that includes textile production, for example, we identify what would happen if enslaved labor disappeared while holding fixed at their actual states everything not on this causal pathway. This includes holding fixed the inactivity of preempted backups. There are no cotton textiles without slave-grown cotton if we hold fixed the fact that other cotton was not in fact shipped to these textile mills. In this way the estimation of a contributory effect will attribute the value of final goods or services produced via some supply chain to each of the chain's causally necessary inputs, where "necessary" means necessary along a single causal pathway.[52] And this will be repeated for each causal pathway—for example, if a commodity is used as an input for more than one final good or service. This is not the same as double-counting, and so does not vindicate it; but it is closer to double-counting than is value-added accounting. Indeed the component effects along a supply chain can sum to more than the value of its final output, but that is not an objection, despite what Olmstead and Rhode suggest. (See the discussion of the pie fallacy in Section 4.1.1.)

In the course of criticizing double-counting, Majewski writes:

> [A]ny commodity can appear critical in a dynamic capitalist economy. Modern economies, after all, are intricate systems of interdependence, so a shortage of a given commodity could seemingly bring the entire system to a grinding halt. Capitalism, however, is also a system of signals and feedback, transmitted via the price system, that allows participants to make adjustments (however painful in the short term) when shortages appear.

The last sentence of this passage need not challenge the first two. We can recognize two different kinds of "criticality," contributory and total.

[51] Note the *separate* reference to "value added" farther down on p. 322.

[52] Lewis (2000: 195): "I think I am doing what historians do. They trace causal chains, and, without more ado, they conclude that what comes at the end of the chain was caused by what went before. If they did not, they could say little about historical causation; because, over intervals of any length, historical counterfactuals become so very speculative that nothing much can be known about the dependence of any event on its causal ancestors." It is interesting to compare, in this connection, the "persistence" literature within (*inter alia*) economic history (Acharya et al. 2024) to the "cluelessness" objections to consequentialism (Lenman 2000) and contractualism (de Kenessey 2025).

I find it helpful to read the New History with this in mind. Laying bare the causal connections between enslaved labor and far-flung corners of the American economy is presumably not enough to establish a claim about total criticality—about counterfactual dependence—because of the forces of equilibration that Majewski highlights. But it might be enough to establish, or at least make plausible, a claim about contributory criticality: the more economic activity that is downstream of enslaved labor, the more plausible that removing this labor *while holding fixed everything not on a single causal pathway* will "bring the entire system to a grinding halt." This is one sense in which enslaved labor could have been more important than suggested by the fact that the products of that labor were only twelve to thirteen percent of antebellum gross national product.[53]

To end this section, I return to the way(s) in which, according to the cliometricians, slavery slowed growth. For all that the cliometricians have said, it is possible that the positive contributory effects of slavery outweighed its negative contributory effects, even if the total effect was negative. That is indeed what we should expect if enslaved labor was "critical," in a contributory sense. This sounds paradoxical but is not: total effect can be negative even if positive contributory effects outweigh negative contributory effects when the former but not the latter are produced redundantly. For example, if everyone who used early birth control pills had also used a preempted backup form of contraception, like "Plan B," then the total effect of the pills would have been to increase the risk of thrombosis even if the prevention of pregnancy—for which the pills were actually responsible—reduced the risk of thrombosis more than the pills increased it. (For more precise versions of this and several of the preceding points, see the Appendix.)[54]

[53] Rhode 2024.

[54] I do not mean to imply that the entire slavery debate is thus dissolved. (Thanks to Taylor Jaworski for pressing me to clarify this.) One especially stark point of disagreement remains whether increases in cotton productivity are attributable to innovations in torture based on an individual quota system, as Baptist claims, or to innovations in biology, per Olmstead and Rhode 2008a, 2008b, 2011. Baptist's main argument against the biological innovation claim is that the evidence for it is better explained by the individual quota system (Baptist 2016: 314–15, n. 36, and see 2014: 127 and 2016: 42–3), while the proponents of the biological innovation claim argue that this system literally did not exist (Olmstead and Rhode 2018: 9–11; cf. Rosenthal 2016: 74).

3 Back to the Simple Observation

I now want to bring the previous two sections together. I will: (a) explain that the simple observation, from Section 1, is only plausible if interpreted in terms of contributory causation, rather than in terms of counterfactual dependence or total causation; (b) argue that this is not an objection to but rather the source of its normative significance; and (c) say what I take its normative significance to be.

One terminological note. From now on I will go back to referring to contributory causation as simply "causation" (and likewise, *mutatis mutandis*, for its cognates) and counterfactual dependence or total causation as, simply, "counterfactual dependence." (I will make the connection back to contributory and total effect when it seems clarifying.) I do this both for economy of expression and because this is the regimentation that philosophers will find more familiar. Readers who accept that counterfactual dependence is its own kind of causation can read that into the discussion.

3.1 The Simple Observation Is an Observation about Causation

In the previous section, Section 2, I argued that the debate between the New Historians and the cliometricians is partly dissolved by the distinction between causation and counterfactual dependence. In characterizing this distinction, I noted that replaceability is relevant to the latter but not the former.

Earlier, in Section 1, I claimed that, if the global economy's low-wage workers stopped working, output would fall to zero, and likewise for its high-wage workers. This is the simple observation.

In making this observation I was relying on an unstated presupposition. Here it is made explicit: if low-wage workers or high-wage workers stopped working *and were not replaced*, then output would fall to zero. The relevant notion of replacement is capacious: recall that we hold inactive all alternative, preempted pathways to economic output. And note how this short-circuits the question of whether the simple observation is about the short or long term. This distinction is important, when it is, in part because rearrangement and replacement take time. But here, in order to identify causation, we are stipulating rearrangement and replacement away.

It is the barring of replacement that drives the result that the low- and high-wage workers contribute equally. To see this, consider Robert Nozick's suggestion, well known to political philosophers, that low-wage workers in an idealized market economy would do worse on their own than high-wage workers, and in that sense gain more from cooperation.[55] Nozick's suggestion seems to amount to the claim that the low-wage workers contribute less than the high-wage workers: if the high-wage workers do better in the absence of the low-wage workers than vice versa, then the high-wage workers make a bigger difference to output:

(actual output per capita − counterfactual output per capita without high-wage workers)

>

(actual output per capita − counterfactual output per capita without low-wage workers).

But Nozick is talking about what would happen if one of these groups left the cooperative venture. The simple observation, by contrast, is about what would happen if they were to leave *and their labor were not replaced*. It is this difference that explains why the latter, but not the former, is about causal contribution, and why the latter, but not the former, leads to an egalitarian result.[56]

3.2 Why Care about Causal Contribution?

Call the difference that a group makes to output its counterfactual contribution. Call the difference that it makes to output (roughly) holding fixed the

[55] Nozick 1974: 193–4. Nozick actually talks about the "better endowed" and "worse endowed," but the basic idea is the same when considering an idealized market economy; see the first footnote on p. 194.

[56] Nozick's suggestion is offered as a criticism of John Rawls's reciprocity-based defense of his *difference principle*. But the general framework of that suggestion is a familiar one. It is basically a bargaining framework. Beyond Rawls himself, where it makes a somewhat muted appearance, it shows up in many places: for example, in David Gauthier's Hobbesian contractualism (1986), in Thomas Nagel's contractualist account of political legitimacy (1991), and in John Roemer's game-theoretic conception of exploitation (1982, 1996, 2017). To be clear, a bargaining framework probably should be cashed out with simple counterfactuals; but what this highlights is that such a framework comes apart from a framework of causal contribution. Thanks to Daniel Viehoff for pressing me to clarify this.

inactivity of preempted replacements its causal contribution. (These are total and contributory effect on output, respectively.) The difference between the simple observation and Nozick's claim illustrates the fact, already operative in the discussion of the debate about slavery, that a group's counterfactual and causal contributions can differ dramatically.

So the simple observation is an observation about causal contributions; and it is because contribution is understood in causal terms that the simple observation has its egalitarian cast, with each group contributing equally.

But why care about the simple observation? Why care about what groups causally contribute? Is our intuitive attraction to the principle of reward according to contribution an attraction to the principle of reward according to causal contribution? The distinction between causal and counterfactual contribution invites these questions, and it does not on its own answer them.

An initial piece of evidence that we care about causal contribution comes from the slavery debate, which is part of the reason I introduced it. Consider the claim that among slavery's evils is the fact that enslaved people were denied the value of the fruits of their labor: as W. E. B. Du Bois puts it, "Great and significant as was the contribution of black labor to the seventeenth, eighteenth, and nineteenth centuries, its compensation approached zero."[57] As we have seen, the claim that slaves' contribution was "great" is, if the cliometricians are right, plausible only when contribution is understood in causal terms. But this is itself evidence that this is how "contribution" ought to be understood in the context of a claim about what enslaved people deserved on the basis of what they contributed. The fact that enslaved workers produced something very valuable but were paid almost nothing beyond the in-kind provision of what was required for their survival seems by itself to show that they did not get what they deserved, a wrong distinct from their being held in bondage and subject to torture.[58] In the same vein, to

[57] Du Bois 2007 (1939).

[58] In one sense the cliometricians do not disagree: they suggest a "rate of exploitation" of at least around fifty percent, meaning that the value of the subsistence goods provided to slaves amounted to no more than fifty percent of the value of their labor (David and Temin 1976: 195–202; Vedder 1975). Since this notion of exploitation uses a slave's individual marginal revenue product (see footnote 8), the rate of exploitation is plausibly lower than if we looked instead at slaves' collective causal contribution. But why would the cliometricians conclude that enslaved labor was exploited at all, if enslaved labor made a negative difference to total output? For one, marginal revenue product is itself closer to a causal notion than a simple counterfactual one: it is the difference between output with and without a worker, *where they are not replaced by a new hire*. I will not say more about

the extent that facts about economic contribution are marshalled to support reparations, it should be facts about causal contribution.[59]

I said that the debate about slavery provides an "initial" piece of evidence. Some philosophers will worry that the real-world case of slavery is too complex to elicit an intuition about the principle of reward according to contribution. Consider, then, this simpler minimal pair:

Tool 1 A worker in a self-sufficient agricultural community makes a tool, only one of which the community needs, using resources equally available to everyone else. The worker is the only one who can make this kind of tool.

Tool 2 As before, but now the worker is one of several who can make this kind of tool. If they do not make it, someone else will.

In both cases, the worker produces a tool. But only in the first case does the worker make a difference that matters. Does this latter fact mean that the worker makes different productive contributions in the two cases, in the sense of "productive contribution" relevant for what they deserve?

To answer this question it is helpful to decompose it into two. First: Is a worker's causal contribution relevant to what they deserve? Second: Is a worker's counterfactual contribution relevant to what they deserve? Four combinations of answers are possible:

		Counterfactual	
		Relevant	Irrelevant
Causal	Relevant	Ecumenism	Productivism
	Irrelevant	Subjunctivism	*n/a*

Productivism is the view that causal contribution matters and counterfactual contribution does not: what matters is who actually produced what. *Subjunctivism* is the converse: what matters is who made what difference.

Tool 1 and Tool 2 are evidence against Subjunctivism. Subjunctivism says that, in Tool 2, the worker does not contribute at all; they do not deserve

this, but it is obviously, in the present context, a pregnant fact.

[59] An appeal to contribution is not necessary for a successful argument for reparations. Nor is it sufficient, since arguments for reparations of the relevant form need both to identify the reparative entitlements of enslaved people and to explain how those entitlements get passed to one's descendants. The principle of reward according to contribution can help only with the first.

anything on the basis of a contribution. (It says the same about enslaved labor in the antebellum South, if the cliometricians' counterfactual conclusion is true.) That seems to me bizarre. If a worker labors to make a tool, and the tool is put to use by their community, it is hard for me to see how it could be true that they contribute nothing, in the sense of "contribute" relevant for what they deserve.[60]

For those who do not share this intuition, let me draw out an odd downstream consequence of Subjunctivism's verdict. Subjunctivism implies that, in Tool 2, *no one* deserves reward for the tool. The main worker does not, because their tool-making does not make a difference; but of course their would-be replacements do not make a difference by making a tool either, since, first, they do not make a tool at all, and, for good measure, even had they made the tool, we can suppose that someone else would have had they not. This, again, seems to me bizarre. How can it be true that a tool has been made but no one has contributed anything? The Contribution Principle is a principle for governing the distribution of what we produce. It would seem unfit for its function if it rendered the verdict that much of what has been produced—everything produced in an average, "replacement-level" way—does not actually count as a productive contribution.

My own intuitive reaction to this minimal pair leads me to Productivism: my reaction is that the worker makes the same contribution in Tool 1 and 2. What they contribute is the value of the tool, where that value is itself understood in causal terms—a function of the causal consequences of what is done with it. (In introducing the notion of a worker's marginal revenue product, at the beginning of Section 1, I mentioned in particular the consequences for preference satisfaction. But for present purposes one need not take a stand on which causal consequences matter.)

Given the implausibility of Subjunctivism, the main rival to Productivism is Ecumenism, according to which both causal and counterfactual contributions matter. There will be many versions of Ecumenism. To say that both causal and counterfactual contributions matter is not to say how much, or

[60] Economists, thinking of *opportunity cost*, might say: Even if the worker in Tool 2 pushes their would-be replacements into idleness, rather than into another valuable job, leisure is itself valuable. So the worker contributes more than nothing, even according to Subjunctivism. First, it is contingent whether leisure is in fact valuable for whoever is preempted. Second, what I think is intuitive is the claim that the worker contributes via the value of what they produce, not via the value of producing it—that is, the value of relieving another of toil.

in what ways, they matter. We might, for example, sum the two kinds of contribution, or instead take their product.[61] I am doubtful that our intuitions are fine-grained enough to adjudicate between the various possibilities, and in any case the differences do not matter here. We need only work with directional verdicts.

For example: Unlike Subjunctivism, Ecumenism does not say that the worker contributes nothing in Tool 2. It says merely that they contribute less than in Tool 1. In Tool 1, the worker makes a positive causal contribution and a positive counterfactual contribution. In Tool 2, they make the same causal contribution but no counterfactual contribution. So any sensible version of Ecumenism will render the verdict that the worker contributes less in Tool 2 than in Tool 1. That verdict still seems to me counterintuitive, but in my experience not everyone shares that reaction; and at a minimum it is less counterintuitive than the verdict rendered by Subjunctivism.

Ecumenism seems to gain more positive support from cases like this one:

Tool 3 As before, but this time the worker who makes the tool knows that they will make a worse tool than those whom they preempt, and in doing so will use up the resources needed to make it, so that no superior replacement can be made.

Many of us will have the intuition that the worker in Tool 3 deserves less than the worker in Tool 1 or 2. Ecumenism offers a straightforward explanation of that intuition: In Tool 3, the worker's counterfactual contribution is negative.

But an alternative explanation of this intuition is available, one that makes use of only causal facts. In cases of preemption, the preemptor is causally responsible for the outcome of interest (the bottle's shattering, the tool's assembly), but they are also causally responsible for the preemption itself—the inactivity of the preempted backup. In some cases, this preemption is itself morally wrong. And it is plausible that this undercuts the person's desert claim—a claim that, on the present view, they would have had on the basis of their causally contributing the outcome of interest. For example: It is presumably wrong to deprive an expert tool-maker of tool-making materials when your community is in need of a tool and would benefit from its being expertly made. According to this explanation, the reason for which the worker in Tool 3 is less deserving is not that their making of the tool amounted to a smaller productive contribution but that there was something defective

[61] In the manner of Hurka 2003.

about how they came to make the contribution that they did—because of what else they caused along the way.

To be clear, Ecumenism need not deny this alternative explanation. Ecumenism agrees that causal contribution matters, and this alternative explanation simply highlights another kind of causal contribution. But the alternative explanation nonetheless weakens the abductive case for Ecumenism, because it shows that we do not need to invoke Ecumenism to explain our intuitive reaction to Tool 3.

We have reached something of a standoff. Let me review. Our reaction to the original minimal pair, Tool 1 and 2, is evidence against Subjunctivism. This leaves us with Productivism and Ecumenism. In my own view Ecumenism, like Subjunctivism, generates a counterintuitive verdict in Tool 2, but as I acknowledged not everyone will agree. Ecumenism seems to gain support from our reaction to cases like Tool 3, but in fact that reaction can be explained in wholly causal, productivist terms.

If we take a step back from this case-based abductive reasoning, though, we find a theoretical consideration that tells in favor of Productivism. Why do we distinguish causation from counterfactual dependence in the first place? Part of the answer seems to be precisely our concern to make backward-looking judgments about responsibility and deservingness. In a forward-looking context, when deciding what to do, we tend to care about what will or will not happen if we do or do not act, which is to say that we tend to care about counterfactual dependence.[62] But in a backward-looking context, when rendering judgments of responsibility and deservingness, we seem to care about who caused what; this seems to be part of what our notion of causation is *for*.[63] In this light, Ecumenism appears unstable. It acknowledges

[62] Hitchcock 2015: 306–7, Hitchcock 2013. This is the basis of the recent debate about whether what has been called "causal" decision theory is really "counterfactual" decision theory. Ibid.; Hedden 2023; Gallow 2024. I do not think of the claim that "we tend to care about what will or will not happen" as incompatible with the central interventionist thesis that the point of causal knowledge is to furnish us with what we need to *intervene* in the world. We often identify counterfactual dependencies *via* causal reasoning: Suzy knows that her rock can cause the bottle to shatter, but also that her not throwing will cause Billy to throw his rock, which can itself cause the bottle to shatter.

[63] Recent work in social psychology, from Sarah A. Wu and Tobias Gerstenberg (2024), might be thought to challenge this received view, or at least the orthodox distinction between causation and counterfactual dependence on which it relies. Wu and Gerstenberg present subjects with several different versions of a case like this: a carpenter, a blacksmith, and a tailor together build a ship; labor of each type is necessary for the construction; but

the normative significance of causal contribution. But if causal contribution is worth distinguishing from merely counterfactual contribution in part for the purpose of making backward-looking judgments of responsibility and deservingness, why would we look to *both* to determine a worker's claim to reward?

I cannot think of an answer. And so for the remainder of the paper I will assume the truth of Productivism: I will assume that the Contribution Principle is to be understood in causal terms. Readers who instead accept Ecumenism can treat the below as explicating only one half of the story about what workers deserve on the basis of their contributions.

3.3 The Egalitarianism of the Simple Observation

What does it actually mean to say that the simple observation has an egalitarian upshot? The low- and high-wage workers together contribute equally. I do not take this to show that: the low-wage workers as a collective deserve economic reward equal to the economic reward that the high-wage workers as a collective deserve, on the basis of both collective contributions being equal. While the collective contributions are equal, these two collectives are not *group agents*. They are just groups—in the jargon, *unstructured* collectives.[64] So they cannot, according to the received view, together deserve anything. Rather I take the simple observation to suggest that: individual low-wage workers deserve economic reward on the basis of what they together do, and individual high-wage workers deserve economic reward on the basis of what they together do, and the rewards in question are equal, because the collective contributions are equal.

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there are different numbers of replacements available for each type. Most subjects assign a worker's labor less causal responsibility for the building of the ship the greater the number of replacements available. Two comments. First, I am unsure what to make of the intuitive judgments collected in these experiments. In the final version, most subjects judge a worker more causally responsible for the outcome when they had "high prior availability" for the work than when they were busy (10–13). This suggests that subjects' responses are tracking some notion of "responsibility" other than causal responsibility. Second, these findings are not reason to abandon the distinction between causal and counterfactual contribution (nor do Wu and Gerstenberg claim otherwise). This is because replaceability has only a small effect on subjects' judgments. That said, any sensitivity to replaceability will, in the present context, narrow the gap between causal and counterfactual contribution. Thanks to Dave Chalmers for telling me about this research.

[64] Nefsky 2019; Collins 2019.

The next section considers a serious challenge to this picture. But its starting point is what I take to be the mainstream approach to unstructured collectives. That approach says that members of an unstructured collective like *polluters* have reasons not to pollute, and can deserve blame for polluting, in ways that are connected to the harms that polluters only together generate. To be clear, this mainstream approach runs up against a problem, what Julia Nefsky calls *the inefficacy problem*.[65] This is the problem of explaining how individuals can have reasons for action connected to, and be rendered deserving on the basis of, an outcome to which they make no significant difference. But there are promising responses to the inefficacy problem.[66] And so I assume that the problem can be solved, such that the mainstream approach, and the features of common-sense morality that it aims to capture, can be vindicated. This is, for my purposes, a crucial assumption: if the problem cannot be solved, then I do not think that anything like the simple observation has clear implications for what workers deserve.

4 Complicating the Simple Observation

The making of the simple observation faces what might seem like a devastating objection. I drew a line down the middle of the global income distribution: low-wage workers on one side, high-wage workers on the other. But we could draw a line anywhere we like. In so doing we could generate different collectives, and, it would seem, thereby generate different conclusions about who deserves what. It is not clear that there is a non-arbitrary way to choose among these different conclusions, and so not clear that we should take any of them seriously. This is an objection, then, not to the simple observation's claim that low- and high-wage workers together contribute equally, but to the idea that this fact is normatively significant—that it can tell us who deserves what.

I agree that we might have no reason to prefer one line over another. But I want to argue that this does not cause the problem that it seems to. This is because the competing delineations do not have to generate different distributive conclusions.

That conclusion, I recognize, is hard to believe. I myself reached it by accident, and with requisite incredulity. I will argue for it in two parts. I first

[65] Nefsky 2019.

[66] E.g., Nefsky 2017 and Lee 2022.

consider distributive conclusions based on different ways of dividing the economy in two. I will then consider conclusions based on dividing the economy into more than two. By "the economy," I continue to have in mind the actual global economy. But what I say applies, like the simple observation itself, to an idealized global economy, or indeed any economy that is comparably complex.

4.1 Different Ways of Dividing the Economy in Two

Imagine a worker who believes that workers like them contribute more and so deserve more than others. What should they say in response to the simple observation?

4.1.1 A Tempting Answer

Here is a tempting answer. They should say that workers like them could by themselves bring the economy grinding to a halt, even though they are relatively few. Suppose, for example, that the worker in question is a dockworker, and that "workers like them" means everyone whose job involves the moving of goods. Call them the "shipping collective." I assume that these workers can, in fact, bring the economy grinding to a halt.[67] The same is presumably true of everyone who is not a dockworker: moving goods is only valuable if there are goods to move. But the shipping collective is a much smaller group. If the shipping collective and the "everyone else collective" contribute equally, such that output should be distributed equally between them, then the individual shipping workers are going to come out with more: their half of the economic pie needs to be distributed among fewer people.

This tempting answer rests on a mistake. What an individual deserves on the basis of being part of a collective that generates some outcome is not directly sensitive to how many other people are members of that collective.

[67] Recall that in making this claim we are imagining that these workers down tools *and are not replaced*. This is not the same possible world that partly determines their bargaining power; that is something like the *closest* world in which they down tools, which may well be one in which they are replaced, at least eventually. This is the specification of the "no agreement point." But the two worlds can be similar, and indeed part of the point of labor unions is to make them similar: withholding labor all together makes replacement—"crossing the picket line"—more difficult, bringing the answer to the question "What would happen if they stopped working?" closer to the answer to the question "What would happen if they stopped working and were not replaced?"

To think otherwise is to commit what Alex Kaiserman calls the *pie fallacy*, "the fallacy of thinking that there is a fixed amount of responsibility for every outcome, to be distributed among all those, if any, who are responsible for it."[68]

Consider Kaiserman's version of a familiar example: two assassins each shoot a victim, who dies; neither shot makes a difference—it is a case of symmetric overdetermination—but each was sufficient to kill the victim.[69] Because each shot was sufficient, there is one sense in which each assassin is wholly responsible for the outcome, and so deserves blame commensurate with its badness. This is to say that they may each deserve what they would have deserved had they been the only one to act. The presence of the other does not *eo ipso* diminish their responsibility.

Now return to the shipping collective. The members of that collective each deserve economic reward commensurate with, and on the basis of, their degree of responsibility for what they together do; and likewise for each member of the everyone else collective. We cannot conclude from the mere fact that there are fewer shipping workers that each is more responsible for what they together do. That would be true if there were a fixed amount of responsibility to go around, but there is not. Of course there is a fixed amount of *output* to be distributed on the basis of this responsibility, but that does not show that the members of the shipping collective are each more responsible for output than the others.

4.1.2 Better Answers That Still Do Not Work

What we need to consider instead are ways to assess degrees of responsibility directly, that is, rather than using group size as a proxy. Philosophers have proposed several of these: they have offered different measures of degrees of causal responsibility. In his survey of this literature, Kaiserman (2018) divides these measures into those that descend from the tradition of thinking of causation in terms of necessity and difference-making and those that descend from the tradition of thinking of causation in terms of sufficiency. Following Hall's (2004) influential discussion, he terms these "dependence" and "production" measures, respectively.

There is one difficulty that needs to be addressed at the outset. These measures are measures of *causal* responsibility. But in looking at whether

[68] Kaiserman 2021: 3598. See also Parfit 1988: 31; Northcott 2013: 3101.
[69] Kaiserman 2021: 3603.

some members of different collectives are more important than others, we are looking at the contribution that a worker makes *to their own collective's contribution*. It is not plausible that a worker's labor is a partial *cause* of their own collective's partly causing output—not if we assume, as is standard, that causes need to be distinct from their effects. It thus makes more sense to think of the worker's labor as a partial *ground* of their collective's contribution, where grounding is an asymmetric metaphysical dependence relation, "the strictest form of the in-virtue-of" relation.[70] I do not want to make anything of this difficulty. I am happy to grant for the sake of argument that some partial grounds are more important than others, and that we can measure the importance of partial grounds in the same way that we measure the importance of partial causes. So in what follows I will, where appropriate, translate talk of causation into talk of grounding, without comment. And sometimes I will talk simply of "importance" or "contribution," letting context disambiguate.

I argue that no dependence or production measures can clearly vindicate

Unequal Responsibility In some cases where two collectives make the same contribution, some members of those collectives are more important partial grounds of their collective's contribution than others.

And thus none can clearly vindicate

Unequal Distribution On the basis of being more important partial grounds of their collective's contribution, some workers deserve greater economic reward.

I will look at three kinds of dependence and production measures.

The first and simplest is a dependence measure. It says that the importance of a partial cause is given by the difference that it makes to what it partly causes.[71] Since I have just spent time emphasizing the distinction between causation and counterfactual dependence—mere difference-making—let me clarify that "the difference that it makes to what it partly causes" here means the difference that it makes when (roughly) holding preempted backups fixed, i.e., contributory effect.

[70] Fine 2012: 38.

[71] This measure has been articulated in a series of papers by Robert Northcott: 2005a, 2005b, 2006, 2008a, 2008b, 2013.

Some individual workers certainly make a bigger difference—have a bigger marginal revenue product—than others, and in that sense are more important partial causes of it. But, as I noted in Section 1, I am assuming that we have already taken off the table the idea that workers can claim credit for the size of the differences to output that they individually make.

For this reason, and this is the second point, Unequal Responsibility is not about an individual's importance for total output; it is about their importance for their collective's contribution to total output. For any individual shipping worker, it is plausible that they make some difference to output, even if small. By contrast, for any one of these workers, their membership in the shipping collective will make no difference to the contribution that the collective makes to output. To see this, imagine that the shipping collective goes on strike, and is not replaced. The economy will grind to a halt whether or not any given shipping worker joins the strike. It is in that sense that their membership in the shipping collective makes no difference to the difference that the collective makes.[72] This makes the generation of this collective's contribution a case of overdetermination, even when the generation of total output is not. Where no worker makes a difference, we cannot use this simplest dependence measure to vindicate Unequal Responsibility.

This does not show there is not some other sense in which some work-

[72] Distinguish these two senses in which a member of a collective might make a difference to the collective's contribution. The first is the one that appears in the text: their membership *in the collective* might make a difference. This requires evaluating three possible worlds. First we have the actual world. Next we have a world in which the shipping collective is on strike, and is not replaced. And third we have a world in which the shipping collective is again on strike but this time differently constituted: the single worker in question is not a member of the collective and so does not join the strike. Our question is whether output differs between the second and third worlds. I answered: no. Now for the second sense of making a difference: a worker's membership *in the economy* might make a difference. This requires evaluating four possible worlds—two pairs. In the first pair we have the first and second worlds from above. In the second pair we again compare a no-strike world to a strike world, but this time imagining the worker in question out of the economy altogether. Our question: Is the difference between the worlds in the first pair bigger? That is, would the abdication of the shipping workers make a smaller difference if the worker in question had never been part of the economy? The answer to *this* question, given what I said in the text, is yes: because the worker makes a difference to total output, they thereby make a difference to the contribution of the shipping collective, by increasing the size of the economy that the collective can bring to a halt. I invoke the first rather than second sense of making a difference because the first seems to capture a less derivative way of making a difference to the collective's contribution.

ers' labor is more important than others'. In some cases of overdetermination, some overdeterminers *do* seem to be more important than others; and there are both dependence and production measures designed to capture this impression. These are the measures best positioned to vindicate Unequal Responsibility. I now argue that they, nonetheless, cannot. I will introduce one dependence and one production measure and identify a problem for their application to the economic case. As will become clear, that problem generalizes.

Chockler and Halpern's (2004) dependence measure asks how many other things would have to have gone differently for a given overdeterminer to have made a difference. If a politician wins an election by just a few votes, then only a few things would have to have gone differently for each vote to have been decisive. If the politician instead wins by many votes, then more things would have to have gone differently for each vote to have been decisive. The latter situation is more overdetermined than the former. The voters in the former are, the thought goes, more important partial causes of the outcome.

Braham and van Hees's (2009) production measure is somewhat similar. Take some overdetermined event. Now collect its causes into *minimally sufficient pluralities* of events. A plurality is sufficient if it suffices for the event's occurrence; it is minimally sufficient if none of its sub-pluralities is sufficient. For each cause, count how many such minimally sufficient pluralities it appears in. The greater the number, the more important the cause.

The collective impact literature distinguishes between *threshold* and *non-threshold* cases. As Samuel Lee puts it, "In threshold cases there is some n such that if at least n contributions to a collective action were made then the outcome of that action would be better/worse than if less than n contributions were made."[73] The two measures above both require that we *find a threshold*—that is, that we find a situation in which at least the cause in question makes a difference.[74] This is straightforwardly true of the first, but it is also true of the second, albeit in a somewhat degenerate sense: if a plurality is *minimally* sufficient, then each of its constituents makes a difference of another kind—a difference to the plurality's sufficiency.[75]

[73] Lee 2022: 79.

[74] I intend "situation" to encompass both something like a possible state of affairs, for Chockler and Halpern's measure, and a subplurality of events, for Braham and van Hees's.

[75] This presupposes that we are continuing to understand difference-making in the way that is relevant for causation, i.e., in a way that holds fixed the inactivity of would-be replacements. Importantly I am not making the claim that the constituents of a minimally

Here is what I want to claim: because these measures require that we find a threshold, they cannot be applied to the economic case, and so cannot vindicate Unequal Responsibility.

Consider again the dockworker with whom we began. Now search for a threshold. For Chockler and Halpern this means a situation in which their membership in the shipping collective makes a difference to the collective's contribution. How do we search? By subtracting other workers. Our preliminary assumption is that, just as in the voting case, once we subtract enough of these others a threshold will emerge.

But there is a problem. As we subtract these other workers, before we reach a situation in which the worker in question makes a difference we may reach a situation in which they *cannot do their job at all*. That is because these other workers are among the causal preconditions of the given worker's labor.

Complex economies are doubly interdependent. First, the size of the *contribution* made by a given worker's labor depends on what other workers do. But, second, that labor *itself* depends on what other workers do. You cannot lower shipping containers onto trucks if they were not loaded onto the ship in the first place. Built into the description of the work of "moving containers" is the presupposition that there are containers to move.

In this respect the economic case is helpfully distinguished from an overdetermined vote. There are some interdependencies in a standard voting case: the votes will not be statistically independent; they will, relatedly, share common causes; and whether or not a vote makes a difference will depend on the other votes. But, for all of that, none of the votes will be a cause of any of the other votes: we vote at the same time, in ignorance of one another. The economic case is not like this. In the economic case, there are causal dependence relations *among* the joint causes of output.

This raises the possibility that there is, for a given worker's labor, no situation in which it makes a difference to their collective's contribution—not because their work is useless but because we cannot reach the situation in which they make a difference without depriving their labor of its causal preconditions. This possibility will be realized whenever the following condition is met: the causal preconditions of a given worker's labor are themselves

sufficient plurality make a difference *to the outcome of interest*—here, the larger collective's contribution. By definition, such a claim will be false whenever there is more than one minimally sufficient plurality.

sufficient for the collective's contribution taking its actual value.

Here is a procedure for checking whether that condition is met. First, select a given worker's labor, like stacking shipping containers. Second, identify what other bits of shipping-related labor need to be performed for that labor activity to be performed. Third, collect these other bits of labor together—the causal preconditions—and ask: what would happen to total output in their absence? If the answer is that it would fall to zero—that these other shipping workers can themselves bring the economy grinding to a halt—then they are sufficient for their collective making a contribution equal to total output. And so the given worker cannot make a difference to that contribution: any situation in which their labor is performed is also a situation in which the shipping workers could bring the economy grinding to a halt with or without them—that is, it is always already a case of overdetermination.

An illustration:

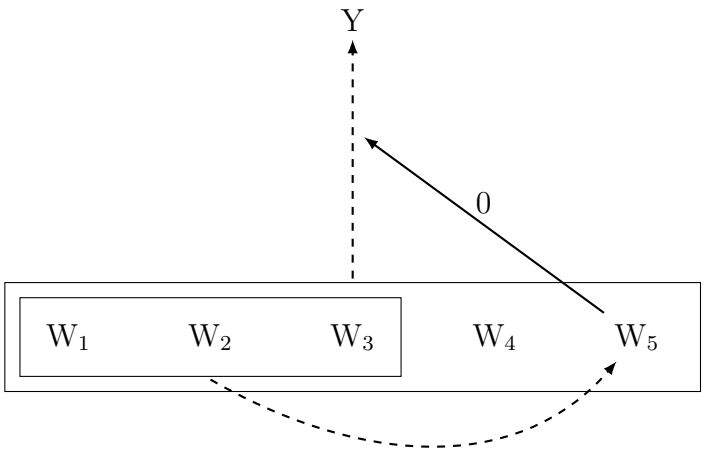


Figure 1: Trying to find a threshold, step 0

The dashed lines represent causal relations. The solid line represents a grounding relation. The dashes lines are drawn from boxes to represent joint causation by what is inside the box: so for example all five workers—W_1 through W_5—are together a cause of output, Y.

Focus on W_5. This worker's labor partly grounds the fact that their collective is a cause of output—the solid arrow. But they make no difference to the difference that their collective makes to output—the "0" attached to that arrow.

Now we want to look for a threshold, to see if we can get W_5's labor to make a difference, turning the "0" into a positive number. First we subtract W_1. Suppose that there is no change:

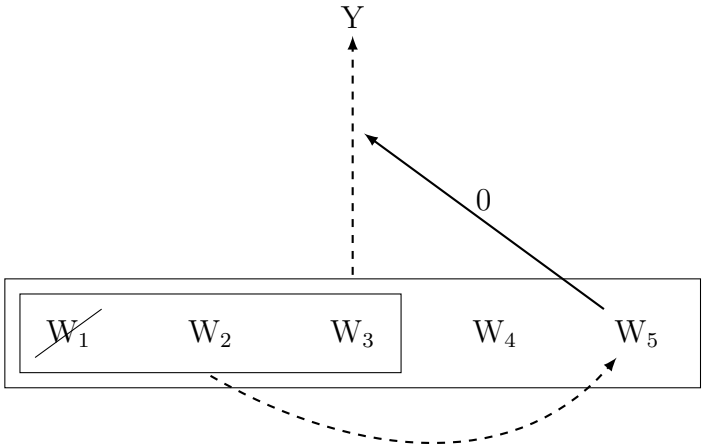


Figure 2: Trying to find a threshold, step 1

Then we additionally subtract W_2. Now suppose that there is a change, but not the one we were looking for. It is not that W_5's labor now makes a difference to their collective's contribution. It is that W_5's labor is no longer performed: W_1, W_2, and W_3 are together a cause of W_5's labor—the curved arrow at the bottom—and without W_1 and W_2, we imagine, W_5 cannot go on:

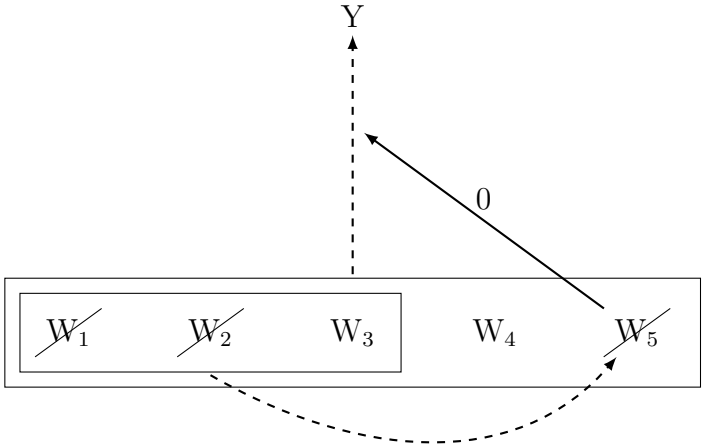


Figure 3: Trying to find a threshold, step 2

The question is this: How often will the causal preconditions of a given

worker's labor themselves be sufficient for the collective's contribution taking its actual value? I am not sure, mainly because I do not know how fine-grained our individuations of labor activity should be. But it seems that, given almost any plausible principle of individuation, the answer will be: very often. Modern economies are astonishingly complex. If one attends to the variety and quantity of goods and services one uses to perform a typical job, and then one imagines out of existence all of the labor (within one's stipulatively defined collective) that supplies those goods and services while holding most everything else fixed, it is difficult to see how one would not thereby imagine an economy brought to a halt. This sort of fragility is what happens when you combine complexity and causal interdependence.

The claim here is not that collective contributions in complex economies are non-threshold cases. Such a claim would face an uphill battle: many philosophers deny that there are any non-threshold cases.[76] My claim is rather that collective contributions are cases of a *disappearing threshold*. Here, in looking for a situation in which a given partial ground makes a difference—a threshold—we end up with a situation in which that partial ground does not even occur.

Thus far I have illustrated the problem of a disappearing threshold within the framework of Chockler and Halpern's measure. What about Braham and van Hees's? The threshold is different, because the difference-making is different, but as far as I can tell we face the same problem, and for the same reason. We will not find minimality if the occurrence of the causal preconditions of a given worker's labor means that we are always already in a situation of overdetermination; if, per the above, these causal preconditions can themselves bring the economy grinding to a halt—if they are sufficient for the collective's contribution taking its actual value—then a plurality that includes them and the labor for which they are preconditions is by definition not minimal. And a plurality that includes only them will not be minimal, either, not if what I have just described is true of an arbitrary worker: each of these causal preconditions has their *own* causal preconditions who are plausibly themselves sufficient. And so on.[77]

If we cannot find a threshold, we cannot vindicate Unequal Respon-

[76] Norcross 1997; Regan 2000; Kagan 2011; and, maybe, Parfit 1984.

[77] This is not a regress that implies the existence of an infinite number of workers. Some bit of labor can be a causal precondition for its own causal precondition, as long as we understand labor as diachronic, so that causation remains asymmetric. This is *inter*dependence.

SIBILITY using anything like the dependence or production measures under consideration.[78]

Let me consider an objection. Even if it is true that the causal preconditions of a given worker's labor are sufficient for the collective's contribution taking its actual value, the objection claims that this problem can be solved using one of several maneuvers from discussions of the semantics of counterfactuals and counterfactual theories of causation. Consider, from the latter, *explicitly nonforetracking* counterfactuals. These are counterfactuals of the form: "If [the cause] had not occurred, but [the effect] had occurred anyway, then..."[79] It might seem that we could use explicitly nonforetracking counterfactuals to push these measures through the barrier raised by the causal interdependence just highlighted. We would do so by evaluating counterfactuals with antecedents like: If the causal preconditions of the dockworker's labor had not occurred, but the dockworker had performed their labor anyway, then...

The problem is as follows. We are stipulating that other workers—the causal preconditions—have no effect on a given worker's labor while hunting for one of their effects, namely the effect of rendering the given worker a difference-maker. Complex economies, I have said, are doubly interdependent. Explicitly nonforetracking counterfactuals would only solve our problem by breaking one kind of economic dependence—the causal dependence of a worker's labor on other workers' labor—while maintaining a second—the dependence of the difference that this labor makes on these other workers' labor. I am not sure if this is coherent; if it is, I cannot see what would justify it. Similar remarks apply to the use in this context of other maneuvers that likewise cleave effects from their would-be causes, like the invocation of *small miracles* that some philosophers, following Lewis (1973: 75), think is required to make sense of our ordinary use of counterfactuals.

[78] I do not mean to imply that, in any disappearing or apparent non-threshold case, we cannot show that some partial causes (grounds) contribute more than others *in any sense of contribution*. The point is that we cannot show that some causes (grounds) contribute more than others on an *output-based* conception of contribution. See footnote 10.

[79] Hitchcock 2001a: 275.

4.2 Dividing the Economy into More than Two Collectives

In the previous subsection we divided the economy into the shipping collective and everyone else. Consider now a division of the economy into more than two groups. We might, for example, simply cleave off from the everyone else collective a group of workers who together make a difference much smaller than the value of total output. The workers who make wool sweaters and the workers who make Ford Broncos make a difference to output, and indeed a difference that is probably larger than traditional "value-added" calculations would suggest (per Section 2.4), but they could not bring the economy grinding to a halt. And likewise for any proper subset of these two groups,[80] and for members of any number of other professions. Being useful without being a *sine qua non* is the normal order of things.

But then the normal order of things can also mislead. Any group that falls short of being itself a *sine qua non* will be *part* of a group that is. In the previous paragraph we began by cleaving off a group from the everyone else collective—a collective that makes a contribution equal to total output. And there is, as far as I can see, no objection to the members of the cleaved off group simply ignoring that cleaving: they are unambiguously members of the everyone else collective, and they can legitimately claim a share of economic output on the basis of *that* membership, rather than another.[81] If what I said in the previous subsection is right, then this share will be an equal one, insofar as we are guided by reward according to contribution. (This is, I assume, not all that we should be guided by.)

To say that these workers have such a claim is not to say that they will make such a claim. This is an important point of connection between the normative claims I am making in this paper and political facts about how workers see themselves. Marxism gives central place to a phenomenon that it helped bring about—the phenomenon of class consciousness. This consciousness is partly consciousness of differences—centrally, in relationships to the means of production. But it is also, the other side of the coin, consciousness of themselves as part of one—very important—collective. The recognition of its importance is recognition of a kind of power, but it is also, on the way of thinking I advance here, recognition of a normative fact. The power is

[80] In saying this I am assuming that each member of these groups makes a difference to output—that none of the jobs are "bullshit" in Graeber's (2018) sense.

[81] Thank you to Samuel Scheffler for first pointing this out to me.

the power to wring out a greater share of output via collective action; the normative fact is that they have a legitimate claim to a greater share, and on the same basis as what grants the power to get it.

4.2.1 The Upshot

Consider a famous claim of Elizabeth Anderson's: "From the point of view of justice, the attempt, independent of moral principles, to credit specific bits of output to specific bits of input by specific individuals represents an arbitrary cut in the causal web that in fact makes everyone's productive contribution dependent on what everyone else is doing."[82]

This claim, while compelling, is not obviously true. Why does crediting each bit of input with the difference that it makes to output represent an arbitrary cut in the causal web? Michael Jordan, to use her example, was helped by the people who kept the court clean, but how many of us really take this to show that there is no important sense in which Jordan contributed more to his team's bottom line (or whatever) than any given ball boy?

Anderson has more recently offered an evocative metaphor for an economy characterized by the kind of interdependence she has in mind: a mechanical clock. Summarizing a 1627 sermon of Robert Sanderson's, she writes:

> God has instituted different callings in society because all are needed to work together in their distinct offices to promote his purposes, like the different parts of a clock. Sanderson preaches that "[T]here is no member in the body so mean or small, but hath its proper faculty, function, and use, whereby it becomes useful to the whole body, and helpful to its fellow-members."[83]

Focus on the idea that each gear in a mechanical clock is essential to its functioning. Now consider two questions. First, is such a clock really analogous to an economy? Second, would this show that each gear "contributed" equally?

On the first question: There is almost nothing that I can bring grinding to a halt, let alone the entire economy. The same seems true of every other individual worker. The metaphor is only apt if we consider collectives of workers, e.g., all the occupants of each role in the division of labor.

[82] Anderson 1999: 321.
[83] Anderson 2023: 11.

On the second question: Even if each gear is essential to a clock's functioning, some gears will be harder to replace, and in that sense make a bigger difference. Thus the metaphor only delivers its intended egalitarianism if we understand contribution in causal terms, as opposed to simple counterfactual ones.

Of course these two conditions—that we consider collectives of workers, and that we understand their contribution in causal terms—will look familiar. The first condition is the one I assumed at the outset of this paper. The second condition is the one that I have used argued for.

This metaphor allows for a more intuitive presentation of the argument of this final section. If you find yourself a part of a "gear" that is necessary for the economy's functioning, that is all well and good, but the rest of the mechanism is necessary, too. And there is no sense in which some individual bit of a given gear is more important than others. You might try to focus in on some little piece of a gear and observe that that piece, unlike the gear as a whole, is not critical; but if you can claim to be part of a gear, rather than part of a piece of one, then so can others.

Of course there will be edge cases, where one suspects that a given bit of labor is so useless it shouldn't even count as part of the economy. These cases are important but are not my concern here. My concern is with inequalities between workers who are, we all agree, part of the economy, part of the clock. For these workers, all that we can and should say is that they are doing their part. Thus, for these workers, a commitment to rewarding contribution is a commitment to equal reward.

A Appendix

I offer a structural equations model of slavery's effect on the nineteenth-century American economy that captures the cliometricians' main counterfactual claims. I then show that, on any of the most popular counterfactual theories of causation, this model vindicates the claim that enslaved people were a cause of America's prosperity.

For a comparison of the causal modeling approach used below and the potential outcomes approach that is more common in economics, see Imbens 2020.

The model that I offer is designed to represent three counterfactual dependence claims:

1. American cotton production did not depend on slavery.

2. New England's cotton textile manufacturing did not depend on domestic cotton production.[84]

3. The South's level of investment in physical capital, by contrast, did depend on slavery.[85]

I start with a narrative description of the historical sequence that is here modeled, in which I identify the states of affairs represented by the model's variables, and the model's assignment of values to those variables. As is typical, these values are discrete, and a value of 0 represents non-occurrence. Not all of the variables are binary, though. Where a variable can be assigned more than two values, a higher number means more of what the variable represents.

After the Revolutionary War, the United States retains the institution of **chattel slavery** ($CS = 1$). So **enslaved labor** ($SL = 1$), not **free labor** ($FL = 0$), **produces domestic cotton** ($DC = 1$) before the Civil War.[86] Cotton is also **produced internationally** ($IC = 1$), but it is not **imported** into New England ($IM = 0$); instead New England uses domestic cotton for the textile production that facilitates **industrialization** ($I = 1$). At the same time, slavery reduces the South's investment in **physical capital** ($PC = 0$). This lowers the country's **growth rate**, even as Northern industrialization and the production of cotton increase it ($G = 4$).

We have two exogenous variables:

[84] We could instead model the claim that the North's industrialization more generally did not depend on textile manufacturing; it would not make a difference. The IM variable below can be given a schematic interpretation as any would-be mechanism for industrializing that was preempted by the actual domestic cotton production and textile production causal chain.

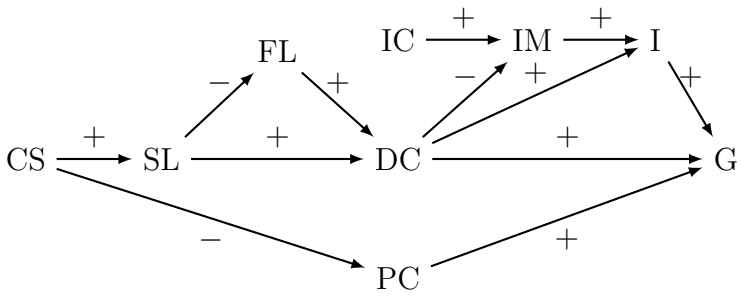
[85] We could instead model one of the other claimed mechanisms by which slavery slowed growth; it would not make a difference. The PC variable can be given a schematic interpretation as any mechanism for slowing growth that is causally independent of the production of cotton and cotton textiles.

[86] Free labor in fact produced around five to ten percent of American cotton even as late as the 1850s (Olmstead and Rhode 2011: 195). I ignore this.

$CS \coloneqq 1$, $IC \coloneqq 1$

The rest are endogenous:

$FL \coloneqq 2 - 2(SL)$, $SL \coloneqq CS$
$DC \coloneqq SL + FL$, $PC \coloneqq 2 - 2(CS)$
$IM \coloneqq IC \wedge (DC = 0)$
$I \coloneqq (DC + IM) \geq 1$[87]
$G \coloneqq DC + 3(I) + PC$[88]



Let me first explain how this model represents the above counterfactual dependence claims. I will then explain how it vindicates the claim that enslaved labor was a cause of American prosperity.

The first counterfactual dependence claim was that American cotton production did not depend on slavery. The model represents this via the relationship between CS, FL, and DC. If America had abolished slavery at its founding ($CS = 0$), then free labor would have produced cotton. In particular, there would have been more free laborers producing cotton than there were enslaved laborers: $FL \coloneqq 2 - 2(SL) = 2 - 2(0) = 2$. And, in the model, domestic cotton production simply requires that there be either free laborers or enslaved people producing cotton. Thus the model captures the claim

[87] Since this is non-standard, this means: $I = 1$ iff DC and IM sum to one or more; otherwise $I = 0$. One could also use the simpler structural equation $I \coloneqq DC + IM$, but this simpler equation embeds the assumption that more cotton means more industrialization, which is not obviously true.

[88] I assume that growth is a positive function of cotton production itself, that is, even holding the production of cotton textiles fixed. In other words, I do not assume that cotton production merely increases the permanent level of output without increasing its growth rate. In making this assumption I am relying on the evidence that cotton productivity increased in the nineteenth century (see note 54). None of the present conclusions is affected by this assumption.

that, without slavery, domestic cotton still would have been produced.

The second counterfactual dependence claim was that New England's textile manufacturing did not depend on domestic cotton production. The model represents this via the relationship between DC, IC, IM, and I. International cotton, the model assumes, is being produced: the value of that exogenous variable is 1. And it would be imported if there were no domestic cotton: $IM := IC \wedge (DC = 0)$. That would, per the model, be sufficient for cotton textile production, because that production requires only some cotton or other: $I := (DC + IM) \geq 1$. Thus the model captures the claim that, without domestic cotton, New England still would have produced cotton textiles.

The third counterfactual dependence claim was that the South's level of investment in physical capital depended on slavery. The model represents this straightforwardly: to get the value of the physical capital variable, PC, we subtract twice the value of the slavery variable, CS. If slavery is abolished—if $CS = 0$—the value of the physical capital variable is larger.

These together imply the further counterfactual claim that slavery was bad for the American economy. Without slavery, (more) cotton would have been produced ($DC = 2$), and the North would have industrialized anyway ($I = 1$), but the South would have invested more in physical capital ($PC = 2$). So growth would have been higher ($G = 7$). Slavery created value redundantly and disvalue non-redundantly.

I now explain why the model vindicates the claim that enslaved labor was a cause of America's prosperity. We begin with the simple theory that Weslake calls (PRE), for "preliminary."[89] According to (PRE) one thing causes a second iff the latter depends on the former holding the "off-path" variables fixed at their *actual* values. Let us apply (PRE) to the model above. $SL = 1$ is our candidate cause; $G = 4$ is our effect. First we hold fixed the off-path variables at their actual values. Then we see whether the value of G depends on the value of SL. And indeed it does: if enslaved people did not produce cotton, then, holding the off-path variables fixed, America's growth rate would have been lower. So, per (PRE), $SL = 1$ is a cause of $G = 4$. CS is also a cause, by the same procedure.

Now, (PRE) is not a plausible theory of causation: it faces immediate counterexamples. (This is why it is "preliminary.") But its main problem is the generation of false negatives: it says that some things which are clearly

[89] Weslake 2015: 7–8.

causes are not. Thus other, more plausible structural equations theories of causation add but do not subtract: everything that (PRE) deems a cause is also deemed a cause by these other theories. But (PRE), we have just seen, deems the labor of enslaved people a cause of America's prosperity. Thus we can conclude that the other, more plausible theories do the same.

Note further that, in the model, the positive contributory effects of slavery outweigh its negative contributory effects, though its total effect is negative.

I end with a nagging worry. In a case of Pearl's (2000: 324), electricity is running through a circuit. If a switch is flipped, electricity will run to Lamp 1, and the room will be illuminated. If the switch is not flipped, electricity will run to Lamp 2, and the room will be illuminated. The switch is flipped, and Lamp 1 turns on. Call this case Switch. Is flipping the switch a cause of the room being illuminated? It doesn't seem like it. (PRE) deems otherwise, and so do the theories that improve on it; they treat Switch like a case of preemption, because it is, in fact, structurally isomorphic to one. Thus the "problem of counterfactual isomorphs."[90] This problem is normally solved by adding to one's theory a *default/deviant* distinction, where causes must be themselves deviant and transmit deviancy to their effects: "while switches affect the *route* by which deviancy is transmitted to an effect, they do not themselves transmit deviancy to the effect."[91]

Is CS a switch? One that in effect turns on a lamp less bright than the one that would have turned on otherwise? We might say: cotton was going to be produced no matter what; the men who failed to abolish slavery at the Constitutional Convention only ensured that it would be produced by workers held in bondage rather than workers who were free.

For CS to have the structural signature of a switch we would need to remove the endogenous variable SL, but that might be okay: it is inessential, in Gallow's sense.[92] The deeper issue is that it is not clear how to apply the default/deviant distinction to this situation. This is a disquieting fact: one wants it to be easier to apply our best theories of causation to the field of social action. But the narrow point is that there seems to be room for reasonable disagreement about CS, in a way that there is not for SL.

[90] Hall 2007: 44ff; Hitchcock 2007; Menzies 2017.
[91] Gallow 2021: 83.
[92] Gallow 2021: 61–2.

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